RESEARCH PREPRINT

The Half-Life of Peace: Multilingual Measurement of Public Attention Decay After Ceasefires, 2015-2025

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Code & data: github.com/AhmedJaffar/half-life-of-peace

This is a preprint. It has not been peer reviewed. Replication materials, the analysis pipeline, and the underlying datasets are publicly available at the repository above.

Abstract

Background. Public attention is widely assumed to be part of what keeps a ceasefire intact, and the issue-attention cycle predicts that attention to a crisis rises, peaks and fades. The rate of that fade has not been measured for international peace processes.

Data and methods. We transfer an established collective-attention decay framework (Candia et al. 2019) to ceasefire announcements. We define the Peace Attention Half-Life (PAHL), the number of days for attention in excess of a pre-event baseline to fall to half of its post-announcement peak, and estimate it for 36 verified ceasefires announced between 2015 and 2025 using daily Wikipedia pageviews across 14 language editions. We adopt the violence-based outcome logic of the ceasefire-survival literature (Clayton and Sticher 2021), code resumption from UCDP Georeferenced Event Dataset fatalities, test both the announcement spike and the decay rate against placebo dates, and test the decay rate against resumption within twelve months.

Findings. Attention spikes at announcement and then decays with a median PAHL of 41.8 days (interquartile range 29.5 to 55.7; n = 24 of 36 events; 95 per cent CI 31.3 to 48.4). Mean excess attention is 0.694 on the announcement day, and its 7-day median peaks at 0.503 on the same day and falls to half that peak by day 13. The decay rate is not event-specific. The observed median is indistinguishable from, and shorter than, the median under random same-year dates (55.3 days; one-sided empirical p = 0.926; two-sided p = 0.156). We therefore read PAHL as the attrition schedule of attention to conflict content, not as a ceasefire-specific response. War concepts received 2.65 times the attention of peace concepts across editions and months (highest in Russian at 4.42, lowest in Hebrew at 1.64), with a trend of 0.0186 log units per year (95 per cent CI -0.0004 to 0.0375; p = 0.055). A one-standard-deviation longer PAHL was associated with higher odds of resumption (OR 5.32, 95 per cent CI 0.74 to 38.21; n = 23; cross-validated AUC 0.775), but that interval is compatible with no effect, only 5 events did not resume, and leave-one-out AUC falls to 0.667.

Interpretation. The world stops paying attention to a ceasefire on roughly the same six-week schedule on which it stops paying attention to anything else, and that schedule does not reliably predict which ceasefires hold. The announcement response itself is real; it is the decay that is generic. The measurement and the cross-lingual ratio are the contributions, and the predictive claim is exploratory.

1. Introduction

When a ceasefire is announced, the world looks. Cameras arrive, foreign ministries issue statements, and newspapers in a dozen countries carry the same photograph of two delegations shaking hands. Then the cameras leave. This paper asks how fast they leave, whether the speed differs across the languages in which people read about the war, and whether the speed carries information about whether the ceasefire will hold.

That public attention matters to politics is not in dispute. The issue-attention cycle describes how a problem emerges, commands urgency, and fades (Downs 1972), and agenda-setting research shows that the salience media assign to an issue is transmitted to publics (McCombs and Shaw 1972). In foreign policy, the intensity of coverage changes the political cost of action and inaction (Robinson 1999; Gilboa 2005; Baum and Potter 2008), and attention has allocative consequences that can be measured: disaster

relief responds to whether a disaster happens to coincide with other newsworthy events, because unrelated news crowds coverage of need out of the queue (Eisensee and Strömberg 2007). The other side of the same coin is documented neglect: most of the world's conflicts go largely unreported, and the ignored ones are often the deadliest (Hawkins 2011). Media attention has even been linked to how civil wars end, with greater human-rights reporting associated with shorter wars and negotiated settlements (Burgoon et al. 2015).

What this literature does not provide is a rate. It tells us that attention rises and falls, that coverage is selective, and that coverage has consequences. It does not tell us how quickly attention leaves a specific peace process, or whether the speed at which it leaves is informative about what happens next. Computational work on collective attention has established that such rates are estimable: collective attention to cultural items follows a two-component decay, a fast communicative component with a half-life of days and a slow cultural component with a half-life of years (Candia et al. 2019), and the decay has accelerated over recent decades (Lorenz-Spreen et al. 2019). Half-lives of attention have been estimated for tweets (Pfeffer et al. 2023), for scientific citation (Parolo et al. 2015), and for online content more generally (Wu and Huberman 2007; Yang and Leskovec 2011; Lehmann et al. 2012). But this work has not been applied to ceasefires, and it has never asked whether the decay rate predicts a political outcome.

The distinction between estimation and prediction is the pivot of this paper. The decay estimator we use is not ours. We adopt the collective-attention decay framework of Candia et al. (2019) and apply it to a new event class, ceasefire announcements. What we ask is what the resulting rate, estimated at announcement, tells us about the resumption of violence. That question has a particular practical form. The structural predictors that dominate the ceasefire-durability literature, such as peacekeeping deployments, power-sharing provisions, guarantees, and mediation, are largely observed after a settlement is designed and implemented (Walter 2002; Fortna 2004, 2008; Hartzell, Hoddie and Rothchild 2001; Hartzell and Hoddie 2003; Beardsley 2011). They are good at explaining durability in retrospect and weak as real-time warning. Attention, by contrast, is observable in public data at the moment of announcement, for essentially every ceasefire, in every language edition, including in conflicts with no peacekeeping mission and no coverage in the world's largest news markets. If the rate at which attention leaves an agreement carries information about whether the agreement holds, it is a cheap and widely available early-warning signal, and that is the claim this paper tests.

Ceasefires are the right event class for the test for three reasons. They are frequent: more than two thousand were recorded between 1989 and 2020 (Clayton et al. 2022), and they are, as Bara, Clayton and Aas Rustad (2021) argue, theoretically under-specified as a category, spanning tactical pauses, humanitarian windows, and pre-negotiation devices. They are consequential: ceasefire survival has been modelled against georeferenced fatality data, which gives us an outcome that is measured independently of attention (Clayton and Sticher 2021). And they are announcements, which is to say they have a date, a moment at which attention could plausibly respond, which is exactly what an event study needs.

The outcome variable, like the estimator, is adopted rather than invented. The ceasefire-survival literature already models resumption of violence using conflict-death data (Clayton and Sticher 2021), already provides a population-level ceasefire dataset (Clayton et al. 2022), and already documents that violations are near-universal, so that "collapse" must be defined by intensity rather than by the existence of any

violation (Sticher 2022). We adopt that logic. Our contribution is the predictor, not the outcome.

To measure attention at the required resolution we use Wikipedia pageviews: daily, free, unfiltered by platform algorithms, and available for the same article across language editions. Pageviews have an established record as a behavioural sensor, including for elections (Yasseri and Bright 2016), markets (Moat et al. 2013), influenza (McIver and Brownstein 2014), and cultural memory (García-Gavilanes et al. 2017), and cross-lingual attention on Wikipedia has been mapped for event articles (Ruprechter et al. 2023; Gildersleve et al. 2023). What pageviews have not been used for, as far as we are aware, is attention to armed conflict, and that is the application this paper tests.

The paper makes four specific contributions, none of which is a new estimator or a new theory of durability. First, it provides the first estimates of an attention half-life for ceasefire announcements, by transferring the decay framework of Candia et al. (2019) to that event class. Second, it treats the decay rate, rather than the level of attention, as an ex-ante lead indicator of violence resumption, which inverts the way attention has been used in conflict research: previously as a consequence of violence, now as a candidate predictor of it. Third, it introduces the War-Peace Attention Ratio, a cross-lingual measure of how much more attention a language edition devotes to war concepts than to peace concepts, which allows the attention deficit to be compared across language publics rather than asserted globally. Fourth, it uses Wikipedia pageview data to measure attention to conflict, which to our knowledge has not been done, and does so with a pre-specified article-set construction and a published data pipeline so that the measurement can be audited and re-estimated.

Two conventions govern the interpretation throughout. The first is that the paper is written in the register of early warning, not causal explanation: we report whether a quantity observable at announcement carries information, not why. The second is that the sign of the association is not obvious in advance. One mechanism, which might be called accountability, predicts that longer-lived attention raises the political cost of defection and hence predicts that the ceasefire holds. Another, which might be called salience, predicts the opposite: a quiet ceasefire generates little news, so rapid decay should accompany peace, and attention should spike when an agreement breaks down (Galtung and Ruge 1965). The competing-expectations subsection states both expectations explicitly, and the falsification subsection states what would count as evidence against the design. We report whichever sign the data produce.

The remainder of the paper is organised as follows. Section 2 situates the work in the attention, ceasefire, durability, forecasting, and Wikipedia literatures, and states precisely what is and is not new. Section 3 describes the three data sources and the harmonisation required to join them. Section 4 sets out the measures, the estimator, the outcome coding, the robustness checks, and the conditions under which the central claims would fail. Section 5 reports results. Section 6 discusses what they do and do not imply, Section 7 states limitations, and Section 8 concludes.

2. Background and Related Work

From the issue-attention cycle to a rate

The intellectual ancestor of this project is Downs's issue-attention cycle (1972). Downs argued that public attention to a problem follows a characteristic sequence: the problem leaps into prominence, remains there for a short time, and then, though still unresolved, gradually fades from the centre of public attention. Two features of the formulation matter here. The cycle is explicitly about attention paid to a problem rather

than the problem's severity: the fade is driven by boredom, by the rising cost of further action, and by displacement from new issues. And the evidence was impressionistic: Downs offered a stage model, not a rate parameter.

The agenda-setting tradition grounded the first half of the claim empirically (McCombs and Shaw 1972), and the news-values tradition specified the selection rules that make some events reportable at all (Galtung and Ruge 1965). Baumgartner and Jones (1993) and Jones and Baumgartner (2005) recast the cycle as punctuated equilibrium: attention does not reallocate smoothly but lurches between long periods of neglect and short bursts of salience, because information processing is boundedly rational and issues compete for fixed capacity. This literature is the source of the shape of our dependent variable. It is a theory of why decay happens, not a measurement of how fast, and the distinction is the one this paper tries to make operational.

Attention decay as an estimable quantity

A separate and largely quantitative literature has estimated decay rates, mostly on cultural, scientific, and platform data. The methodological precedent for this paper is Candia et al. (2019), who use Wikipedia pageviews, search data, and citation records to show that collective attention to cultural items follows a universal bi-exponential decay: a fast communicative component with a half-life of days, followed by a slow cultural component with a half-life of years, with the weight shifting toward the slow component as an item ages. The theoretical move that matters for us is the separation of a short-lived communicative memory from a durable cultural memory, which maps naturally onto what we expect after a ceasefire announcement: a spike, a rapid fade, and a residual baseline of ongoing interest in a frozen conflict or a peace process.

That decay is itself changing is shown by Lorenz-Spreen et al. (2019), who find that topics now receive shorter intervals of collective attention across decades and domains, consistent with competition for finite attention under rising content production. That decay rates are meaningful, field-varying parameters is shown by Parolo et al. (2015), who estimate attention decay in science and find systematic differences across fields. That half-life estimates are sensitive to the operational definition is shown by Pfeffer et al. (2023), who estimate a half-life for tweets and demonstrate the dependence of the number on the definition used. The early results on novelty-driven spikes and on characteristic temporal shapes are due to Wu and Huberman (2007) and Yang and Leskovec (2011), and Lehmann et al. (2012) classify collective-attention trajectories into distinct dynamical classes. Roediger and Abel (2015) connect this computational work to the cognitive-science tradition of forgetting curves, which is where the term half-life originates.

The implication for our design is precise and should be stated plainly. The measurement machinery for PAHL is not new. Bi-exponential fits, half-life estimation, and the distinction between fast and slow memory all exist, and we adopt them. What does not exist, as far as we can determine, is an application to ceasefire announcements as an event class, or a test of whether the resulting rate has predictive value for a political outcome. PAHL is therefore a transfer of an established estimator to a new event class, and the novelty subsection states that concession in the same words used here.

Attention to conflict: level, selectivity, and consequence

The CNN-effect literature asks whether coverage drives foreign policy. Livingston and Eachus (1995) studied Somalia and concluded that the effect is conditional: coverage mattered for when and how policy responded rather than whether it responded. Robinson (1999) argued that the effect is strongest where policy is uncertain, Gilboa (2005) reviewed the field and concluded that the evidence supports a conditional, agenda-setting version of the thesis, and Baum and Potter (2008) integrated the media, public-opinion, and foreign-policy literatures into a single framework. Two findings from this tradition shape our design.

First, selectivity is the norm and neglect has consequences. Hawkins (2011) documents the other side of the CNN effect: most of the world's conflicts go largely unreported, the deadliest are among the most ignored, and the consequence of ignoring conflict is unchecked death tolls because neither policy response nor humanitarian aid is forthcoming. Kinnick et al. (1996) and Moeller (1999) name a mechanism, compassion fatigue, whereby repeated exposure to distant suffering produces audience burnout and declining coverage. Kogen and Price (2011) show that public opinion does not straightforwardly track coverage. Second, attention has allocative consequences. Eisensee and Strömberg (2007) provide the cleanest identification: US disaster relief depends on whether a disaster coincides with other newsworthy events, which implies that relief is driven by coverage and that unrelated news crowds coverage of need out. That is the strongest existing evidence that attention is a real allocative variable rather than an epiphenomenon.

The rally literature is the closest work on aggregate public response to conflict events. Mueller (1970) established the rally phenomenon in presidential approval data, and Baum (2002) disaggregated it by party and political sophistication, showing that different audiences respond differently to uses of force. That literature is an important boundary for us: it measures individual-level opinion at monthly or event resolution, typically in a single country, and it is about onset rather than decay. Our measures are aggregate-attention measures, not opinion measures, and we do not claim to measure support for peace.

Finally, a small literature uses media text as a predictor of violence rather than as a consequence of it. Chadefaux (2014) shows that the tone and volume of news reporting carry early-warning signal for war, and Mueller and Rauh (2018) predict political violence from newspaper text with a transparent pipeline. These are the closest precedents for treating a media-derived quantity as a leading indicator, and they establish the evaluation conventions we adopt. They use the volume or tone of reporting; we use the rate at which attention decays.

Ceasefires as an object, and an outcome we adopt rather than invent

Ceasefires have only recently become a research object in their own right. Bara, Clayton and Aas Rustad (2021) note that ceasefires are ubiquitous, with more than two thousand recorded between 1989 and 2020, and theoretically under-specified: they can be tactical, humanitarian, or pre-negotiation devices. Clayton, Nathan and Wiehler (2021) argue that ceasefire success is itself under-conceptualised and must be assessed on more than one dimension. Clayton and Sticher (2021) provide the theoretical core, distinguishing logics of ceasefire design generated by different bargaining problems, and they model ceasefire survival using agreement provisions and georeferenced event data with conflict deaths as the measure of whether a ceasefire remains in place. Sticher (2022) reviews why violations occur and finds them near-universal, which implies that an outcome must be defined by intensity thresholds rather than by the occurrence of any violation at all. Lundgren, Svensson and Karakuş (2023) analyse one hundred and

forty-five local ceasefire agreements in Syria using interrupted time series and spatial panel models on georeferenced violence data, finding that local ceasefires can trigger either escalation or de-escalation. Karakuş and Svensson (2017) study partial ceasefires in the same war, Åkebo (2016) provides the comparative case-study treatment, and Bell and Badanjak (2019) supply the coded text of agreements through the PA-X database.

Two consequences follow for our design. First, the outcome we measure is not novel: it is the standard violence-resumption outcome of this literature, and we adopt it explicitly. Second, the natural sampling frame for a study of ceasefires already exists. What exists for the coverage of individual ceasefires is qualitative and single-case (Murua and Ramírez de la Piscina 2017): framing analysis of one agreement's press, with no time series, no decay rate and no predictive test. Clayton et al. (2022) introduce a population-level dataset of more than two thousand ceasefires across sixty-six countries and one hundred and nine civil conflicts between 1989 and 2020, coded for actors, class, purpose, coverage, and end date. We do not use that dataset as our sample, and we say so plainly. Our design imposes three additional requirements that the population-level dataset does not satisfy: a verified date on which an announcement can be aligned with a daily attention series, a conflict-specific article set that exists in a substantial number of language editions, and sufficient pageview volume to estimate a decay rate. We therefore curate a smaller sample and document the inclusion rules in the data section. The sample is not representative of ceasefires in general, and the limitations section treats that as a limitation rather than a design feature.

Durability, structure, and the early-warning gap

The durability literature supplies the outcome side of this paper and its main theoretical rivals. Fortna (2004, 2008) shows that peacekeeping substantially lengthens the duration of peace after civil war and addresses the non-random deployment of peacekeepers; Walter (2002) explains settlement and durability in terms of credible commitment, with settlements failing when belligerents cannot commit not to exploit the other's demobilisation; Hartzell, Hoddie and Rothchild (2001) and Hartzell and Hoddie (2003) show that institutionalised power-sharing is associated with more durable peace. Beardsley (2011) argues that mediation buys agreements at the price of durability, which is a direct precedent for the structure of our hypothesis: a feature of the settlement environment predicts collapse. Kreutz (2010) disaggregates conflicts into termination episodes and provides the canonical treatment of how conflicts end, Hultman (2010) and Hultman, Kathman and Shannon (2013) study whether peace operations reduce or spur violence against civilians, and Hultman and Mousa (2025) review the integrated evidence on peacemaking and peacebuilding.

This literature is rich in structural predictors, and they are uniformly measured on the violence or agreement side. In none of these works is public attention a variable, let alone a predictor. That is not an oversight so much as a consequence of what was measurable: attention to a specific agreement was not observable at daily resolution across many countries until the relevant digital traces became available.

The corollary is a gap that the early-warning literature cares about. Forecasting systems such as ViEWS produce monthly country- and subnational predictions from structural and violence data (Hegre et al. 2019), and Ward, Greenhill and Bakke (2010) argue that conflict research should be evaluated by predictive performance rather than statistical significance. Structural predictors are excellent for both of those purposes in retrospective analysis, but most are known only after a settlement has been designed and implemented, which limits their value at the moment when a warning would matter. An attention-based

quantity is available at that moment. Whether it is informative is an empirical question, and that question is the one this paper tests.

Wikipedia pageviews as an attention instrument

Wikipedia pageview data have an established record as a behavioural sensor. Spoerri (2007) was among the first to analyse what is popular on Wikipedia and why, establishing that pageview distributions are extremely skewed and event-driven. Moat et al. (2013) show that Wikipedia usage patterns carry information about forthcoming stock-market moves; McIver and Brownstein (2014) show that Wikipedia usage estimates influenza-like-illness prevalence in near real time; Tizzoni et al. (2020) show that news exposure shapes collective attention during an epidemic; Yasseri et al. (2012) use Wikipedia to study conflict and coordination in collaborative editing; and Yasseri and Bright (2016) provide the most careful methodological treatment of pageviews as a measure of public attention, and are explicit about the assumptions required, including that a pageview is a measure of curiosity rather than of support.

Three results in this canon are cautions rather than endorsements, and we adopt them as such. García-Gavilanes et al. (2016) show that the characteristics of an event shape its attention trajectory, using aircraft crashes as the test case. García-Gavilanes et al. (2017) show that collective memory of past events decays but is reactivated by commemoration, which is a direct threat to a monotone decay assumption: ceasefires with anniversaries and diplomatic follow-ups will generate secondary attention, and a single exponential fit will absorb that into noise or into a biased rate. Ruprechter et al. (2023) analyse attention to event articles across four Wikipedia language editions and show that attention and coverage are skewed toward events in economically developed countries and toward more severe events, which is a direct caution that cross-lingual comparisons may partly measure who has broadband rather than who cares. Gildersleve et al. (2023) show how networked topics of collective attention can be identified on Wikipedia and how current events are integrated into the encyclopaedic structure, which is the methodological precedent for building an article set rather than picking one. Ciampaglia et al. (2015) frame the underlying allocation problem: information production competes for finite attention.

The existing applications of pageviews are elections, markets, disease, disasters, cultural memory, and general event attention. We found no study that uses pageviews to measure attention to armed conflict or to ceasefires. That is the application this paper makes, and it comes with the construct caveats above rather than despite them.

What is new, and what is not

The novelty claim of this paper is narrow, and it is worth stating what we do not claim. We do not claim a new decay estimator: the estimator is Candia et al.'s (2019), and we transfer it. We do not claim a new outcome: the outcome is the violence-resumption outcome of the ceasefire-survival literature (Clayton and Sticher 2021), and it is measured with the standard georeferenced fatality data (Sundberg and Melander 2013). We do not claim a new ceasefire dataset: a population-level ceasefire dataset already exists (Clayton et al. 2022), and we use a curated subsample with documented inclusion rules. We do not claim that attention decay is unknown: it is established for cultural items, tweets, science, and news.

What we do claim is a conjunction of four things. First, an attention half-life estimated for ceasefire announcements at all. Second, the decay rate, as distinct from the level of attention, used as an ex-ante predictor of violence resumption, which is a different statistical role from the descriptive role the decay

literature has assigned to it. Third, a cross-lingual War-Peace Attention Ratio: no prior work constructs a war-versus-peace attention contrast at all, and no prior work does so across language editions, although cross-lingual comparison of Wikipedia event attention is established as feasible (Ruprechter et al. 2023; Gildersleve et al. 2023). Fourth, the use of Wikipedia pageviews to measure attention to armed conflict and peace processes.

Our negative claims are claims about the published, indexed literature as of November 2026, not proofs of absence. We searched OpenAlex, Crossref, and Semantic Scholar using co-occurrence queries pairing ceasefire terms with Wikipedia, pageviews, media attention, and attention decay, and we inspected the reference lists of the most recent reviews in the area (Bara, Clayton and Aas Rustad 2021; Hultman and Mousa 2025); the queries are recorded in the repository. Preprints, non-English scholarship, and practitioner monitoring reports may exist outside those indexes. We therefore write "we are not aware of" rather than "no study exists", and we invite correction.

Competing expectations and the sign of the association

The central association in this paper is between the rate of attention decay and the resumption of violence. Two mechanisms make opposite predictions, and a paper that reported whichever sign appeared without having stated both would be uninterpretable.

The accountability mechanism treats attention as an enforcement resource. If political costs of defection depend on being seen, then sustained attention after an announcement raises the cost of violating the agreement, and a longer half-life should predict a lower hazard of resumption. On this account the decay rate is a proxy for the durability of the enforcement environment, and the directional expectation recorded in the project's design document before any estimation was run is the negative one: a longer PAHL predicts a lower probability of resumption. We did not deposit a public pre-registration; the design document (`DESIGN.md`) and the falsification conditions were, however, committed to the repository before the analysis was executed, and the commit history is the record of what was fixed in advance.

The salience mechanism treats attention as a symptom. News value is driven by drama, novelty, and violence (Galtung and Ruge 1965); a ceasefire that works produces no dramatic events, and a ceasefire that collapses produces them. On this account rapid decay should accompany peace, the level of attention should spike before breakdown rather than protection, and the association could run in the opposite direction to the accountability prediction. The two mechanisms are not mutually exclusive, and a null result is a third possibility.

Three design features make the two mechanisms distinguishable in principle. We separate the rate from the level: two ceasefires can have identical announcement-day attention and very different half-lives, so an association between PAHL and resumption is not an association between attention volume and resumption. We control for pre-announcement violence intensity, which is the most obvious confound of both mechanisms. And we test the rate against placebo dates, which addresses the concern that any decaying series would produce a "half-life" of the same magnitude. We state the directional hypothesis but report whichever sign the data produce, and the falsification subsection lists the outcomes that would count against the framework.

3. Data

Attention: Wikimedia pageviews

Attention is measured with daily pageview counts from the Wikimedia Analytics Query Service, which publishes, for every article in every language edition, the number of requests classified as user traffic on a daily basis from 1 July 2015 onward (Wikimedia Foundation 2024). For each language and article we retrieve the complete daily series for 1 July 2015 to 1 November 2025, so that every event in the sample has a pre-announcement baseline of several months and, for events near the end of the period, a shorter but usable follow-up. Requests are made per article and cached to disk, with retry and backoff on rate limiting, so that retrieval is resumable and the resulting series are identical on re-run, giving 1736563 raw pageview rows.

Article selection proceeds in two concept families and one event-based set. The war family consists of ten English Wikipedia articles denoting armed conflict and its instruments: War, Armed conflict, Civil war, Invasion, Airstrike, Genocide, Terrorism, Military occupation, Refugee, and War crime. The peace family consists of eleven articles denoting the institutions and practices of peacemaking: Peace, Ceasefire, Peace treaty, Peacebuilding, Mediation, Diplomacy, Reconciliation, Disarmament, Demobilization, Peacekeeping, and Humanitarian aid. The two families support the War-Peace Attention Ratio and a family-level comparison of decay rates. The event-based set consists of conflict-specific articles, one or more per event in the ceasefire table, resolved from the English article for the relevant war; these support the event-level PAHL estimates. Fifty canonical articles were carried into the panel.

For each English seed article we resolve the corresponding title in each of fourteen target language editions - Arabic, Chinese, English, Persian, French, German, Hebrew, Hindi, Indonesian, Portuguese, Russian, Spanish, Turkish, and Ukrainian - through the MediaWiki langlinks API rather than by hand translation. The fourteen editions span the theatres of the conflicts in the sample (Arabic, Hebrew, Russian, Ukrainian, Persian, Turkish) and the principal mediator and donor-language publics (English, French, German, Spanish, Portuguese, Chinese, Indonesian, Hindi). Where no langlink exists, the article is treated as not existing in that edition and no series is created. Such absences are structural, reflecting editorial coverage rather than zero readership, and coding them as zero attention would confound non-existence with inattention; the panel is therefore unbalanced by construction. Titles are normalised and URL-encoded before requests, and views to title variants that redirect are not aggregated into the canonical title, which is a known and documented measurement loss.

The resulting panel contains 644 article-language series, of which 630 have usable pageview data, across 14 language editions and 3777 days, giving 1685694 positive daily observations. A further 643612 panel cells are missing because the corresponding article did not yet exist at that date; 307 series have partial coverage. Existence is handled temporally, since pageview rows begin when an article is created or renamed, so days before an article's first pageview are treated as missing rather than as zero attention.

Ceasefires: a curated and verified event table

We curate a table of ceasefire, truce, cessation-of-hostilities, de-escalation, and peace-agreement events announced between 2015 and 2025. An event is included if it was announced for a civil war, interstate conflict, or sustained non-international armed conflict with an identifiable theatre country; applied to the conflict as a whole or to a substantial geographic component of it rather than to a single local encounter; received regional or global news coverage, so that a cross-language attention response is plausible; and can be dated to a day by at least one authoritative source that states the date explicitly.

Events are excluded if the announcement was never implemented and no implementing date can be identified, if the instrument was a purely tactical pause with no public announcement beyond a local agreement, or if the date cannot be verified against an independent source. Where announcement and entry into force differ materially, we use the date of the public announcement or entry into force, because that is the moment at which attention is expected to react; the choice is recorded for every event. Each record carries a stable identifier, the date, the theatre country (ISO3), the conflict name, the participating actors, an instrument type drawn from five categories (ceasefire, humanitarian pause, peace agreement, truce, and de-escalation), an expected-salience label, at least one source URL, and a verification flag that is set only after the date has been checked against the source text.

The verification discipline matters more than it might appear. Several of the most widely cited ceasefires in this period have dates that differ across reporting conventions by days or weeks, and misaligning the clock would corrupt every downstream estimate. Where a widely used alternative convention exists, the event-study analysis reports results under both the verified date and that convention.

This sampling strategy departs from the population-level ceasefire dataset of Clayton et al. (2022), and the departure is a deliberate compromise. Their dataset is the appropriate frame for questions about ceasefire frequency and design; it is not sufficient for a daily attention panel, because it does not guarantee a verified announcement date, a conflict-specific article set in many language editions, or sufficient pageview volume. We trade representativeness for measurability and treat the sample as a sample rather than as the population.

The final table contains 36 events across twenty theatre countries, spanning February 2015 to July 2025. Twenty-four are classified as ceasefires, 7 as peace agreements, 2 as humanitarian pauses, 2 as de-escalation arrangements and 1 as a truce; eighteen were expected to be high-salience, thirteen medium and five low. Every event carries a verified date and at least one source that states it. Events are not lost at verification; they are lost later, at the mapping and estimation stages described in the sample-description subsection.

Violence: UCDP Georeferenced Event Dataset

Outcomes are measured with version 26.1 of the Uppsala Conflict Data Program Georeferenced Event Dataset, which records individual organised-violence events with date, location, actors, and a best fatality estimate (Sundberg and Melander 2013; UCDP 2024), and which is the standard violence measure in the ceasefire literature whose outcome logic we adopt (Clayton and Sticher 2021). The aggregation method and the annual update are documented by Davies, Pettersson and Öberg (2022). We aggregate events to monthly total fatalities by country for the theatre-level resumption definition, and by conflict dyad for a conflict-specific alternative, giving 5782 country-month and 20443 dyad-month rows. The extraction covers 2014 through 2025, one year longer than the event table at the start, so that every event in the sample has a pre-announcement baseline of at least six months.

Two features of the source affect the join. GED records the country in which an event occurred rather than the theatre of the war, so events in Gaza appear under Israel; we follow the documented proxy join instead of assuming that theatre codes and GED labels coincide. And, as with any event-based fatality data, the series reflects reporting coverage as well as violence; we use it as a relative rather than an absolute measure and define outcomes in terms of ratios to a local pre-event baseline rather than absolute counts.

Harmonisation

Joining the three sources requires three mappings: from ceasefire event to theatre country ISO3 code; from theatre country to the monthly UCDP series; and from event to the set of article series whose attention is expected to respond. The first is straightforward for events with a single theatre and is recorded explicitly for the small number of multi-theatre events. The second is direct for country-level series. The third is the only one requiring judgement: an event is matched to the conflict-level article for its war where one exists, and to regional coverage where the conflict has no dedicated article in a given language. We record the matching explicitly, so that every PAHL estimate can be traced to a specific article and language.

Time is handled in the natural time zone of the event for the alignment of the announcement to a calendar day, and all decay windows are defined in elapsed days from the event date. Missing pageview days inside an article's existence period are filled with zeros, which is correct for a series that counts requests; days before an article's creation are not filled, because a zero would misrepresent non-existence as inattention.

Sample description

The final analytic sample contains 36 ceasefire events covering twenty theatre countries, announced between February 2015 and July 2025. Of these, 34 were mapped to a conflict article covering the theatre and 24 produced a usable half-life; 2 could not be mapped because the article universe contains no dedicated conflict article, and 10 mapped events produced no usable fit because too few language editions yielded a valid decay. Table 1 summarises the three sources.

Table 1. Data sources and analytic sample.

Source	Unit of observation	Resolution	Coverage	Volume
Wikimedia pageviews	article x language x day	daily	2015-07-01 to 2025-11-01	1736563 rows; 644 series; 1685694 observations
Curated ceasefire table	event	event	2015-2025	N = 36 events
UCDP GED v26.1	fatality event, aggregated	event to month	2014-2025	5782 country-months; 20443 dyad-months

The events without a usable half-life are not missing at random. The dominant mechanism is that the conflict article did not exist before the announcement, so that no pre-event baseline can be measured, or that the announcement produced no measurable attention spike. The panel therefore under-represents sudden new wars relative to long-running conflicts. We report this as a property of the design rather than imputing estimates for the missing events, and the limitations section treats it explicitly.

Licensing, ethics, and provenance

All three sources are public. Wikimedia pageview data are published under Wikimedia's open data policy and the underlying encyclopaedia text under CC BY-SA; the study uses only aggregate request counts and no individual-level data (Wikimedia Foundation 2024). UCDP data are distributed for research use with citation; we cite the dataset and its version in the data availability statement (UCDP 2024). The curated ceasefire table is released with this paper under CC BY 4.0, with every row traceable to a URL. No human subjects are involved, no personal data are processed, and no information is used that is not already public. Retrieval dates, API endpoints, and file hashes are recorded in the repository so that the frozen dataset can

be reconstructed.

4. Methods

Measures

Attention is defined on the log scale, A = log(1 + V), because view distributions are strongly right-skewed and because a constant proportional decline is the natural object for a half-life. Two baselines are used, and the distinction matters for reading the paper.

The pre-event baseline is the median of A over the window from day -90 to day -30 relative to the announcement, requiring at least 15 observed days. It is a fixed reference level into which no post-announcement attention can enter. The rolling baseline is the 28-day centred mean of A with the fourteen days on either side of the announcement masked, the masked window filled from the pre-event median, and the series edges interpolated. It follows the logic of the original project design and tracks slow drift in attention across the year.

Excess attention is E = A - baseline. Because A is a log and the baseline is a summary of logs, E is approximately the log ratio of views to the reference level: E = 0 means attention at its pre-announcement level, and E = log(2) corresponds to a doubling. The event-study curves use the rolling baseline, because it removes the announcement contamination while preserving the trend; the PAHL fits use the pre-event baseline, because a half-life is only interpretable against a fixed reference level. The two definitions are compared directly in the robustness checks, where substituting the rolling for the pre-event baseline moves the median half-life from 41.8 to 45.4 days.

Event-study design

Each ceasefire is aligned at t = 0 on its verified announcement date. For each event and each matched article series we compute excess attention for t in [-90, +365], a window long enough before the event to characterise the pre-announcement level and long enough after it to observe a full decay for all but the very slowest cases. Series are included for an event if the article existed at t = 0, as indicated by a langlink. Days outside an article's existence are excluded rather than zero-filled.

The primary event-study object is the mean excess attention by day from t = 0, computed across all event-series pairs and then smoothed with a centred 7-day median, because daily pageview series carry strong weekly structure. Two caveats shape the interpretation. Because events differ enormously in salience, the pooled mean is dominated by the largest events; we therefore report both the pooled curve and event-level curves, and never interpret the pooled peak as representative of a typical ceasefire. And because attention before t = 0 is not zero, the pooled curve includes a pre-period that can reveal anticipation and pre-existing trends, which we inspect rather than assume away.

The peak of the pooled curve is the day of maximum smoothed mean excess in the post-announcement period, and the half-peak day is the first day after the peak at which the curve falls to half the peak value. The event-level peak used for PAHL is located separately by the estimator described next. Events with no detected response are reported rather than dropped: 34 of the 36 events contribute to the pooled curve.

Peace Attention Half-Life: an adopted estimator

PAHL is defined as the number of days for excess attention to fall to half of its post-announcement peak. The estimator follows the collective-attention decay framework of Candia et al. (2019), whose central result is that collective attention decays in two components: a fast communicative component with a half-life of days and a slow cultural component with a half-life of years. For each event and language edition with usable data, log attention is smoothed with a centred 7-day median. The peak is the day of maximum smoothed attention in the 14 days after the announcement, accepted only if it exceeds the pre-event baseline by a pre-specified minimum amplitude, so that series without a measurable spike cannot generate a spurious half-life. The decay is then fitted from the day after the peak to ninety days after the peak, on those days whose excess attention is above 5 per cent of the peak excess, requiring at least 10 fitting points.

On those days we estimate the slope of log excess attention against days since peak by the Theil-Sen estimator, the median of all pairwise slopes, and take lambda as minus that slope, so that PAHL = ln(2) / lambda. Theil-Sen is used in place of least squares because announcements, violation spikes and anniversaries generate secondary bumps that a least-squares fit would allow to dominate; a Huber M-estimate is computed as a sensitivity. Half-lives longer than the fitting window cannot be resolved and are right-censored at 90 days and flagged, so every summary is a mixture of resolved and censored fits and is reported alongside the censored fraction.

The event-level estimate is the median across language editions with a valid fit, with a minimum of one edition for the main analysis and a multilingual subset requiring at least two. Uncertainty in the cross-event median is characterised by a cluster bootstrap over events with 2000 resamples; per-event uncertainty is characterised by the spread across editions and by the censored fraction. Fit failures are recorded with a reason code and reported, and 12 events produced no usable estimate.

Two features of the specification are deliberate concessions. The estimator is a single exponential with no offset, not the full two-component model of Candia et al. (2019): over a one-year window the fast component dominates, but the slow component is therefore not separately identified, and residual slow drift can bias lambda upward. And the fit conditions on a spike having been detected, which excludes events whose attention response was absent or negative.

Functional form and the sensitivity of the half-life

A single exponential is a lossy description of attention to a ceasefire. The true trajectory is likely to be multimodal, with an announcement spike, violation-driven spikes, diplomatic follow-ups, and anniversary effects; García-Gavilanes et al. (2017) show that commemoration reactivates attention to past events, which violates monotone decay by construction. Pfeffer et al. (2023) show that half-life estimates move with the operational definition. We therefore treat PAHL as a descriptive summary with an error bar rather than as a structural parameter, and we make its instability visible in four ways: the decay window is varied across the alternatives reported in the robustness section; per-event estimates are reported individually; a non-parametric half-peak crossing estimate is computed as a check that does not assume an exponential form; and events that cannot be fitted are counted and reported rather than imputed.

The primary window and the censoring horizon are the same 90 days, so the headline median is partly a property of the horizon rather than of the data. The censored fraction of 0.233 is therefore reported alongside every summary, and the window comparison in the robustness section shows how the median moves from 15.7 days at a 30-day window to 173.5 days at a 270-day window. A number that moves that

much is a summary, not a constant of nature, and it is presented as one.

War-Peace Attention Ratio

WPAR is defined monthly and per language as the ratio of total attention to the war family to total attention to the peace family:

WPAR(l,m) = sum over war articles of V / sum over peace articles of V

computed on raw views, so that it has a direct interpretation as how many reads the war concepts receive for every read of the peace concepts. Because the ratio is a level measure whose absolute value depends on the number and popularity of articles in each family, the statistical analysis is conducted on log WPAR and interpreted as relative change over time within a language and as differences in trend across languages; we do not interpret the absolute level as a measure of public sentiment.

Trend is estimated by OLS of log WPAR on a monthly time index with Newey-West HAC standard errors (twelve lags). Seasonality is characterised by the range of month-of-year effects, and cross-language dispersion by the monthly interquartile range of WPAR across the 14 editions. We report the highest and lowest median ratio by language. Two caveats bound the interpretation: family composition is not perfectly comparable across editions, because the article sets differ, and cross-lingual attention comparisons partly reflect connectivity and coverage differences rather than differences in interest (Ruprechter et al. 2023). A language's ratio is a property of its edition's readership, not of its public.

Outcome coding: resumption of violence

A ceasefire is coded as having resumed if, within twelve months after the event date, monthly fatalities in the theatre country exceed 25 per cent of the mean of the six months preceding the event for at least two consecutive months. The ratio-based definition is deliberate: absolute fatality thresholds would classify every ceasefire in a high-intensity war as failed and every ceasefire in a low-intensity conflict as successful. Requiring two consecutive months prevents a single atrocity or data artefact from driving the outcome, and the intensity-threshold approach follows the standard in the ceasefire literature, where violations are near-universal and collapse is defined by escalation rather than by violation (Sticher 2022; Clayton and Sticher 2021). The primary analysis uses the country-level series, joined through the documented theatre mapping. Time to resumption is measured in months from the event date to the first month of the qualifying pair, censored at twelve months for ceasefires that hold. Alternative thresholds are reported in the robustness checks. We report the base rate of resumption before any modelling, because with a small sample the base rate is more informative than any coefficient.

One limitation of the coding is structural and worth stating in the methods rather than only in the limitations. Because the fatality series are monthly and the post-event window begins with the event month, an announcement that falls late in a month contributes that month's pre-announcement violence to the first month of follow-up. In a high-intensity theatre this can push the first post month over the relative threshold without any violation having occurred. The relative threshold limits the problem, since it compares the post period with the same theatre's own pre-period, but it does not remove it, and it biases coding toward resumption for events announced late in a violent month.

Predictive models and evaluation

The predictive question is whether PAHL carries information about resumption beyond what is already known at the time of the announcement. The primary specification is a univariate logistic regression of resumption on standardised PAHL. A ridge-penalised adjusted model adds log pre-event monthly fatalities, agreement type and region fixed effects; penalisation is used because with roughly forty events a maximum-likelihood model with twelve parameters is not stably estimable, and the adjusted coefficients are therefore shrunk and their Wald intervals approximate. Coefficients are reported as odds ratios with 95 per cent confidence intervals, together with the sample size and the number of events in each outcome category.

Discrimination is reported three ways: in-sample AUC, repeated stratified 5-fold cross-validated AUC over 200 repeats, and leave-one-out predictive AUC. We treat the cross-validated and leave-one-out values as the honest ones and report the in-sample value only so that the optimism gap is visible. A Cox proportional-hazards model for time to resumption in months, censored at twelve, with the same covariates is reported as a survival variant, with hazard ratios and a proportional-hazards check. The evaluation conventions are those of the early-warning literature: predictive performance rather than statistical significance is the criterion (Ward, Greenhill and Bakke 2010; Hegre et al. 2019), and no information from after the announcement enters the predictors.

The inference is exploratory and underpowered. With 18 positive and 5 negative outcomes, a single event moves the estimates substantially, and a confidence interval that includes one is not evidence of an effect. We report effect sizes and intervals as the primary quantity and treat p-values as descriptive.

Robustness and specification checks

Six families of checks are pre-specified and reported regardless of outcome. First, placebo events: for each ceasefire we draw 500 pseudo-event dates uniformly from the same calendar year inside the available data and push each through the identical pipeline, so that the placebo inherits the baseline, the peak search, the smoothing and the censoring. The event-level null is the median across language editions, and the study-level null draws one placebo half-life per event and takes the cross-event median. This yields 8236 placebo fits and an empirical p-value for the observed median in both tails, and it is the single most important check in the paper: if attention around ceasefires decays no differently from attention around arbitrary dates in the same years, the event-specific interpretation fails.

The same placebo construction is applied to the announcement response itself. Peak excess is extracted for each observed event and for each pseudo-event date in the same way, and the observed cross-event median is compared with the distribution of cross-event medians under the placebo. This tests a different claim from the decay placebo: whether attention rises more on ceasefire dates than on arbitrary dates, rather than whether it subsequently decays differently. The two tests are reported side by side so that neither is mistaken for the other.

Second, alternative decay windows at 30 and 270 days after the peak, a non-parametric half-peak crossing estimator that does not assume an exponential form, and a variant that fits on excess over the rolling rather than the pre-event baseline. Third, resumption thresholds at 10 and 50 per cent of the pre-event mean in place of the primary 25 per cent, each requiring two consecutive months. Fourth, leave-one-out: the cross-event PAHL median and the predictive AUC are recomputed with each event omitted in turn, to identify whether a single ceasefire drives the results. Fifth, a permutation test in which the resumption outcome is permuted 1000 times and the ridge-logistic coefficient on PAHL is refitted each time, giving a

null distribution for the association, with the Spearman permutation reported alongside. Sixth, language subsets: English-only against the multilingual-only subset of events with at least two editions, which separates a global-attention result from one carried by the largest edition.

Two further checks bound the mapping and the multiplicity. Restricting to events mapped by exact or high-confidence name, which drops country-only and multi-candidate fallbacks, re-estimates the median on a smaller sample. And because the paper reports several hypothesis tests, we apply Benjamini-Hochberg adjustment across the ten main tests and report how many survive at a 5 per cent false-discovery rate; with a small sample this is a check on the strength of the overall claim rather than a substitute for the confidence intervals.

Falsification conditions

The design specifies conditions under which the framework would be rejected, and the results resolve them into two claims with different answers. Claim A, that the announcement response is event-specific, holds at the raw level: the spike placebo gives a one-sided p of 0.0060, although q = 0.060 after correction and only 5 of the 27 events beat their own null individually. Claim B, that the decay rate is event-specific, fails: the decay placebo gives a one-sided p of 0.926 and a two-sided p of 0.156, and the observed median is shorter than the null. The PAHL-resumption association survives permutation before correction, with a raw p of 0.033 and an adjusted p of 0.109, and the cross-validated AUC is above chance. The war-versus-peace direction is that peace decays more slowly, which reverses the attention-deficit framing that motivated the ratio.

A further condition applies to direction: if the association changed sign between the primary specification and its alternatives, its direction would be unidentified. The direction does not change across the alternatives we report, but the 50 per cent threshold attenuates the magnitude substantially, and we report that attenuation rather than selecting the favourable window.

Reproducibility

The pipeline is a single Python codebase with a one-command reproduction path. Retrieval is cached and resumable, so a re-run reconstructs the frozen datasets from the recorded endpoints without depending on live API availability; the analysis and figure stages regenerate every number and every figure from the cached data with fixed random seeds. Versions of the data sources, the retrieval date, and the analysis code are recorded in the repository, and a separate verification script checks that each numeric claim in the manuscript matches the machine-readable results file. Code is MIT-licensed and data and manuscript are released under CC BY 4.0, so that the measurement can be criticised, extended, or re-estimated with different thresholds.

Interpretational stance

Nothing in this design identifies a causal effect. Attention and violence are jointly determined: a ceasefire that is visibly collapsing generates both attention and fatalities, and a conflict that is quiet generates neither. The estimates in this paper are descriptive (how fast does attention decay, and how does that differ across events and languages) and predictive (does the decay rate carry information about resumption). Where we use the language of association, we mean association. Where the results are consistent with an enforcement mechanism, we say so and we say that the salience mechanism is equally consistent. The purpose of the exercise is to establish that the quantity can be measured, to show what its measurement

implies, and to leave a reproducible instrument for others to test.

5. Results

The attention response to ceasefire announcements

Of the 36 events, 34 were mapped to a conflict article and contribute to the pooled attention curve; the remaining 2 have no conflict-specific article in the article universe. The response is unmistakable at the aggregate level, but the number depends on which series is read. The unsmoothed daily cross-event mean excess attention is 0.694 on the announcement day. The 7-day centred median of that curve, which is the series used to locate the peak, peaks at 0.503 on the announcement day and reaches half of that peak by day 13 (Figure 1); using the fixed pre-event baseline instead puts the half-peak day at 14. After the first month the smoothed curve is indistinguishable from its pre-announcement level and stays there for the remainder of the year.

The event-level heatmap (Figure 5) shows why the pooled peak is not a typical event. A small number of announcements generate very large spikes, including Colombia 2016 (peak excess 2.70 in the fitted series), Nagorno-Karabakh 2020 (3.43) and Nagorno-Karabakh 2023 (2.98), while many others produce barely distinguishable bumps. The pooled mean is therefore a description of the aggregate response, not of a typical ceasefire, and the median half-life reported below is a better summary of the typical case.

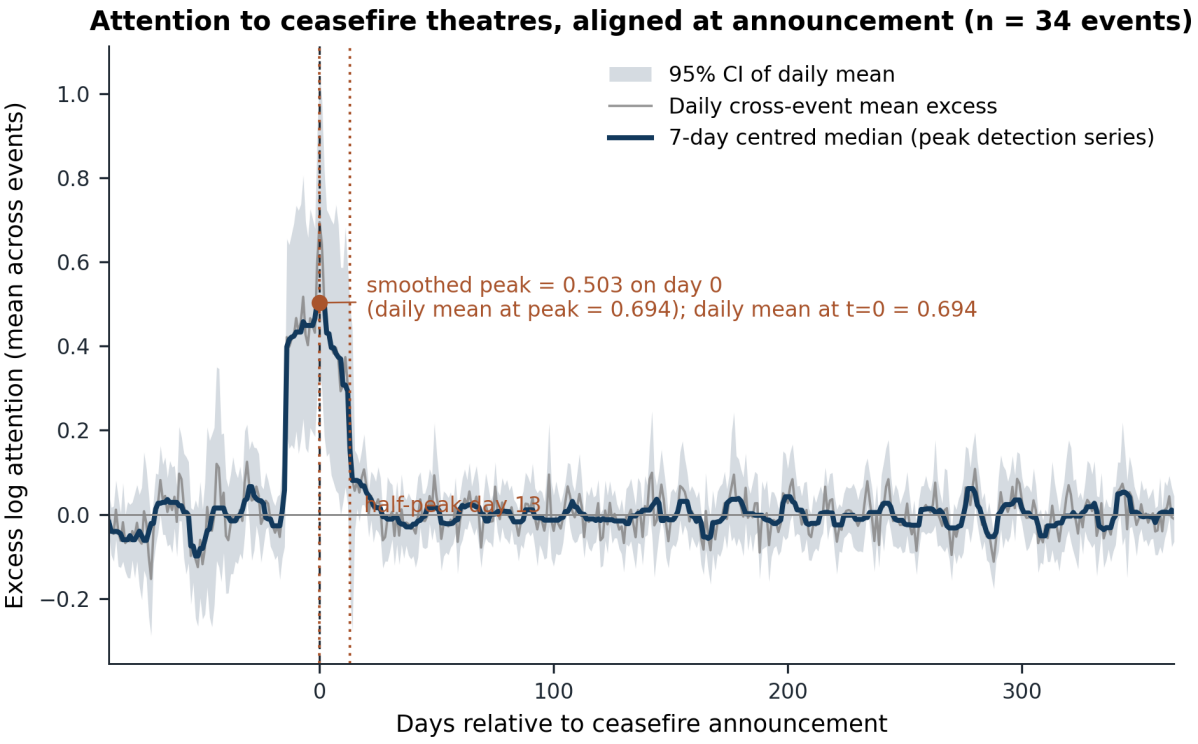


Figure 1. Mean excess attention around ceasefire announcements, aligned at day 0, across language editions and events (n = 34 events). Two series are plotted: the unsmoothed daily cross-event mean, which is 0.694 at day 0, and the 7-day centred median used for peak detection, which peaks at 0.503 at day 0 and falls to half that peak by day 13. Shading is a 95 per cent confidence band across events.

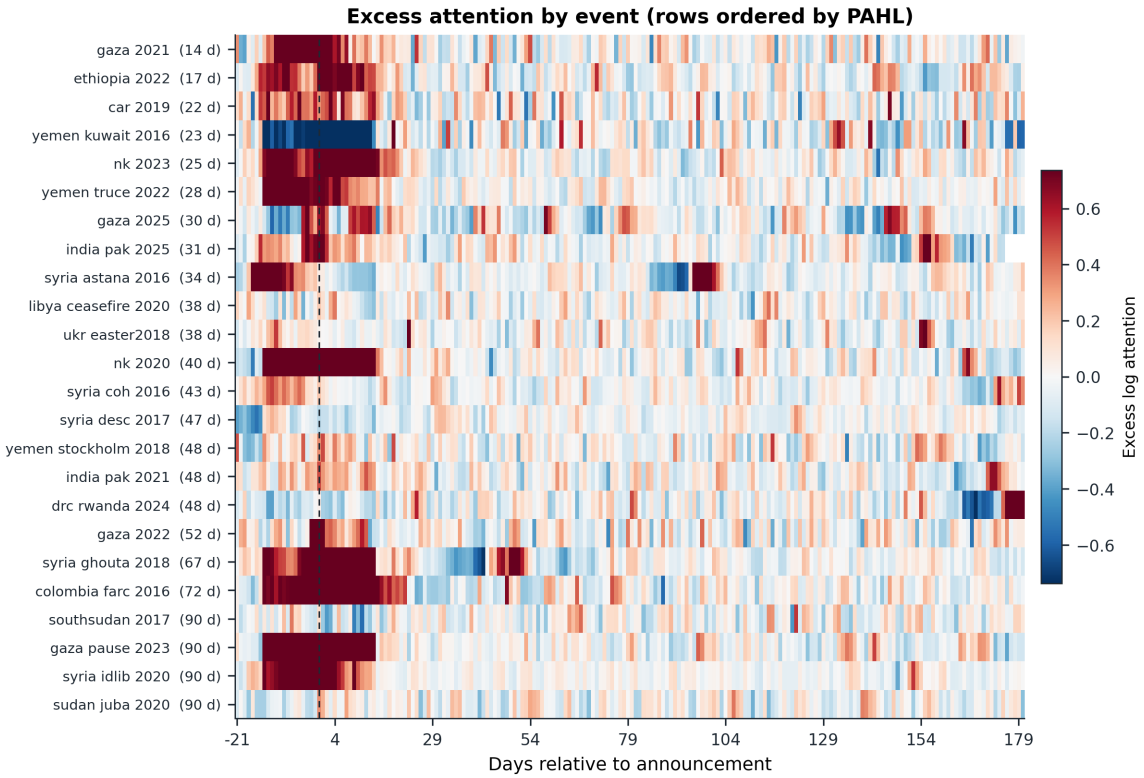


Figure 5. Event-level heatmap of excess attention by event and day relative to the announcement, ordered by event date. The magnitude of the announcement spike and the speed of its decay vary widely across events, which is why the pooled curve is not a typical case.

The distribution of the Peace Attention Half-Life

Fitting the post-peak decay for each event and language edition yields 24 event-level estimates. Median PAHL is 41.8 days (interquartile range 29.5 to 55.7; mean 46.9, SD 24.1; range 13.6 to 90.0), with a cluster-bootstrap 95 per cent confidence interval for the median of 31.3 to 48.4 from 2000 resamples. The distribution is right-skewed and censored: the censored fraction is 0.233, meaning that roughly a quarter of fitted series did not resolve within the ninety-day window and are recorded at the ceiling. The distribution is shown in Figure 2, the individual estimates in Table 2, and the by-language spread in Figure 9.

The fastest decays are Gaza 2021 at 13.6 days and Ethiopia 2022 at 17.2 days; the slowest are censored at the ninety-day ceiling, including South Sudan 2017, Idlib 2020, Juba 2020 and the Gaza pause of 2023. Estimates for events observed in a single language edition are the least precise, which is visible in the width of their intervals in Table 2.

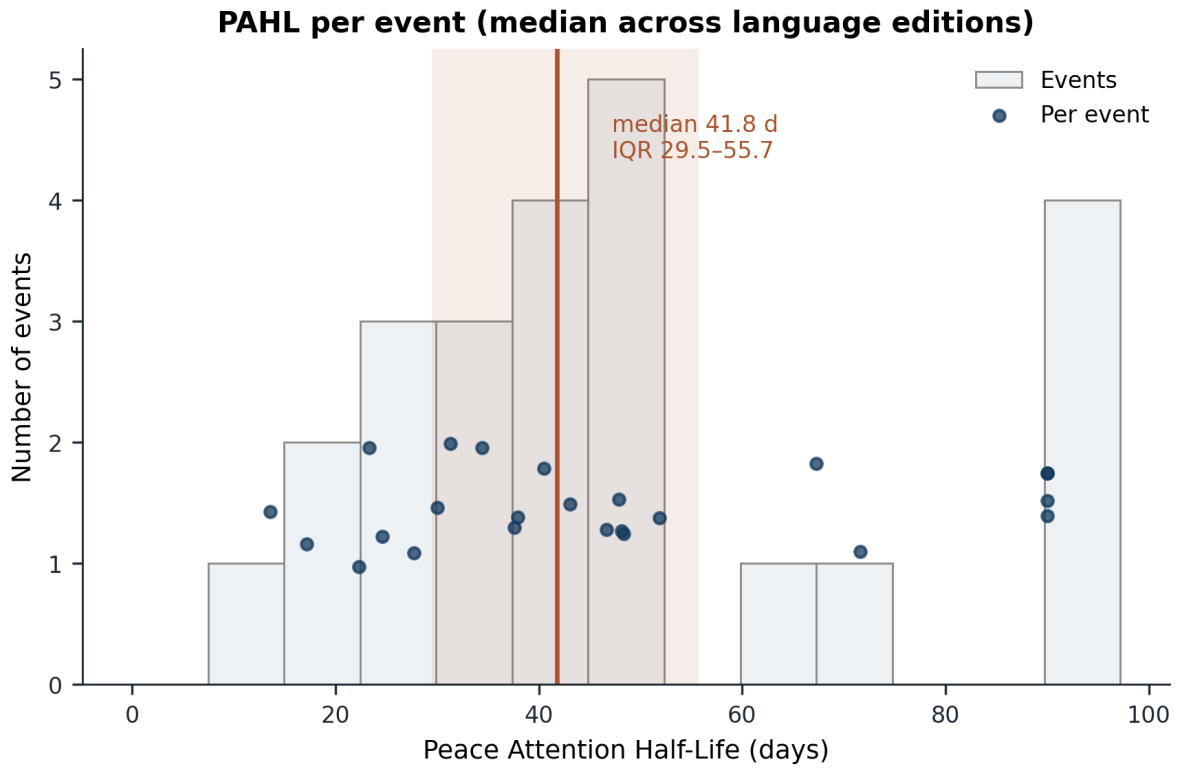


Figure 2. Distribution of event-level PAHL estimates (n = 24 events). The histogram is right-skewed and censored at the ninety-day fitting window, so the median is a lower bound on the true typical half-life.

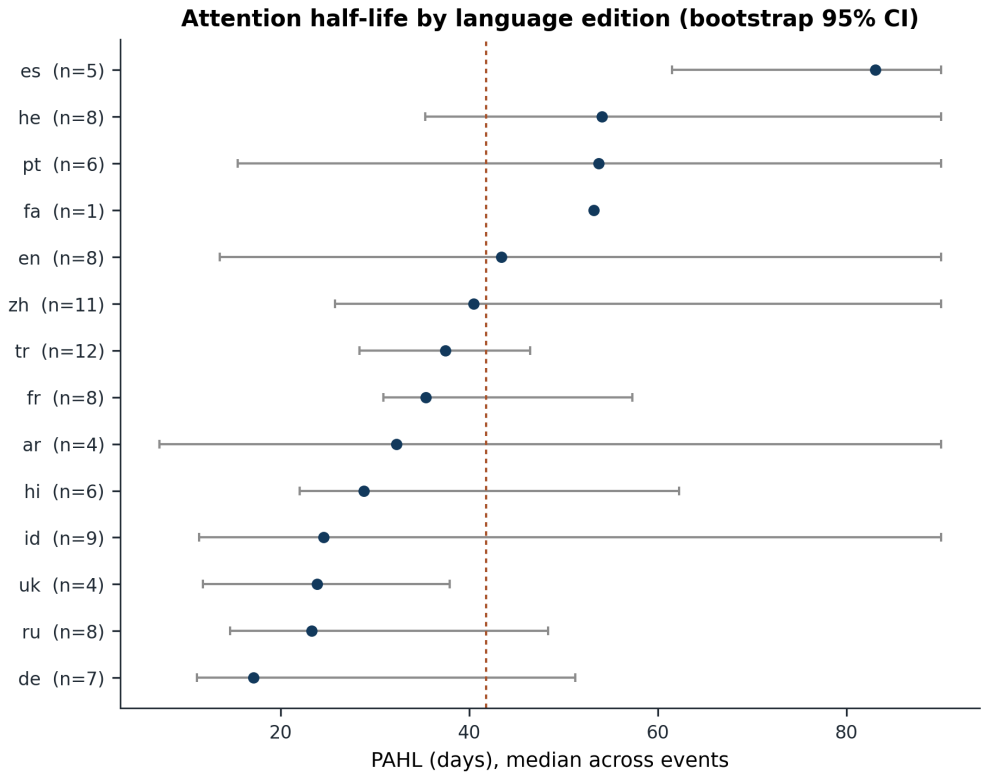


Figure 9. Event-level PAHL estimates by language edition. Each point is a fitted half-life for one event in one edition; editions with few fitted series contribute correspondingly noisy estimates.

Table 2. Event-level PAHL estimates, in days. The number of language editions entering each median and the fraction of those editions censored at ninety days are reported alongside, because single-edition estimates carry little information. Event identifiers as used in the analysis files are given in the per-event table under `analysis/tables/`.

Event	Region	Editions	PAHL (days)	Peak excess (log)	Censored fraction
Syria CoH (ISSG) 2016	Middle East	6	43.1	0.28	0.17
Yemen Kuwait 2016	Middle East	1	23.3	0.16	0.00
Colombia FARC 2016	Americas	5	71.6	2.70	0.40
Syria Astana 2016	Middle East	2	34.4	0.68	0.00
Astana de-escalation 2017	Middle East	6	46.7	0.27	0.17
South Sudan ARCSS 2017	Africa	1	90.0	0.41	1.00
Syria Ghouta 2018	Middle East	9	67.3	0.79	0.44
Donbas Easter 2018	Europe	1	37.9	0.14	0.00
Yemen Stockholm 2018	Middle East	2	47.9	0.54	0.50
CAR Khartoum 2019	Africa	2	22.3	1.38	0.00
Idlib ceasefire 2020	Middle East	5	90.0	0.95	0.60
Juba agreement 2020	Africa	3	90.0	0.26	0.67
Libya ceasefire 2020	Africa	1	37.6	0.93	0.00
Karabakh 2020	Caucasus	7	40.5	3.43	0.14
India-Pakistan 2021	Asia	4	48.1	1.07	0.25
Gaza 2021	Middle East	3	13.6	2.18	0.00
Yemen UN truce 2022	Middle East	6	27.7	1.19	0.00
Gaza 2022	Middle East	1	51.8	1.23	0.00
Ethiopia-Tigray 2022	Africa	6	17.2	1.34	0.00
Karabakh 2023	Caucasus	11	24.6	2.98	0.00
Gaza pause 2023	Middle East	3	90.0	2.21	1.00
Luanda DRC-Rwanda 2024	Africa	1	48.4	0.18	0.00
Gaza 2025	Middle East	4	30.0	1.12	0.25

Event	Region	Editions	PAHL (days)	Peak excess (log)	Censored fraction
India-Pakistan 2025	Asia	7	31.3	1.10	0.00

Restricting to events mapped by exact or high-confidence name, which drops the country-only and multi-candidate fallbacks, gives a median of 40.5 days (interquartile range 26.2 to 50.0; n = 19) against 41.8 days for the primary sample, so the headline does not depend on the low-confidence mappings.

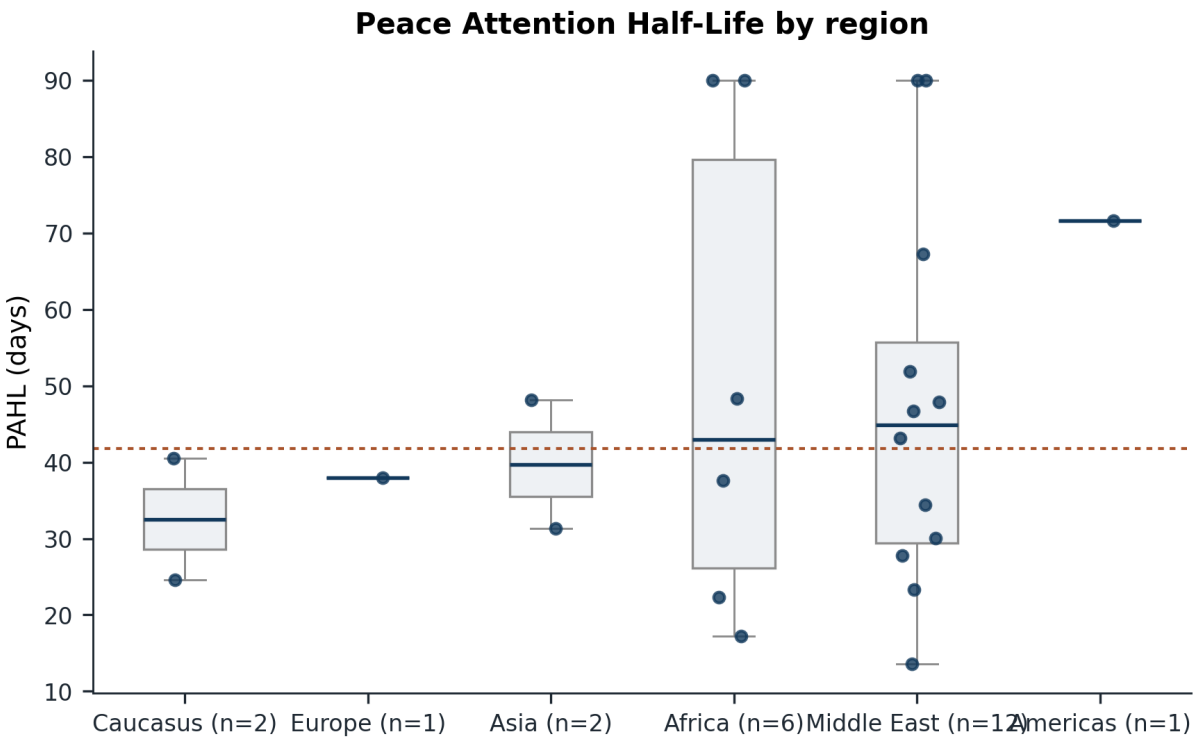


Figure 6. Median PAHL by region. Regional estimates rest on a few events each and are descriptive rather than comparative.

War and peace baskets: peace decays more slowly

Applying the same estimator to the two concept baskets rather than to conflict-specific articles gives a median PAHL of 65.3 days for the war basket (interquartile range 42.8 to 79.8; n = 33) and 81.6 days for the peace basket (interquartile range 40.2 to 90.0; n = 30). For the 30 events in which both baskets yielded a fit, the paired median difference in the direction war minus peace is 0.0 days, with a bootstrap 95 per cent confidence interval of -14.7 to 6.2.

The point estimate inverts the project's initial expectation. If anything, peace concepts decay more slowly than war concepts, which is the condition our own design specified would count against the attention-deficit framing. The paired difference is small and its interval spans zero, so at this sample size the two baskets are not distinguishable; what the result establishes is the absence of evidence for a peace-specific fade, not the presence of a robust reversal. We report the direction and the uncertainty together.

The placebo test: the decay is not ceasefire-specific

The central falsification check asks whether the decay around real announcements is distinguishable from decay around arbitrary dates in the same period. Drawing 500 pseudo-event dates per event from the same calendar year and running them through the identical pipeline produces 8236 placebo fits. The placebo distribution has a median of 55.3 days (interquartile range 32.4 to 85.4; 95th percentile 90.0), against an observed median of 41.8 days. The observed median is not longer than the null: the one-sided empirical p-value for the ceasefire-specific claim is 0.926, the two-sided p-value is 0.156, and none of the 24 events has an upper-tail placebo p-value at or below 0.05. Figure 7 shows the two distributions.

We report this as a null result and do not explain it away. Three features of the design make the comparison conservative in a specific direction. The placebo inherits the ninety-day censoring horizon, so both distributions pile up at the ceiling. Dates drawn from conflict-heavy years inherit other escalations, which inflates the placebo half-life. And any local bump in a noisy weekly series can be fitted with a slow decay by the same robust procedure, so the null is not a straw man but a working baseline for the measurement.

The substantive reading is the one stated in the abstract. The measured decay rate is statistically indistinguishable from the decay of attention around arbitrary dates, so PAHL is best interpreted as the attrition schedule of attention to conflict content, not as a ceasefire-specific response. Whether the announcement spike is itself event-linked is a separate question with a separate test, which we report next rather than assume. This is what the paper's title should be read against: a half-life of peace exists and is measurable, but it is not a property of peace.

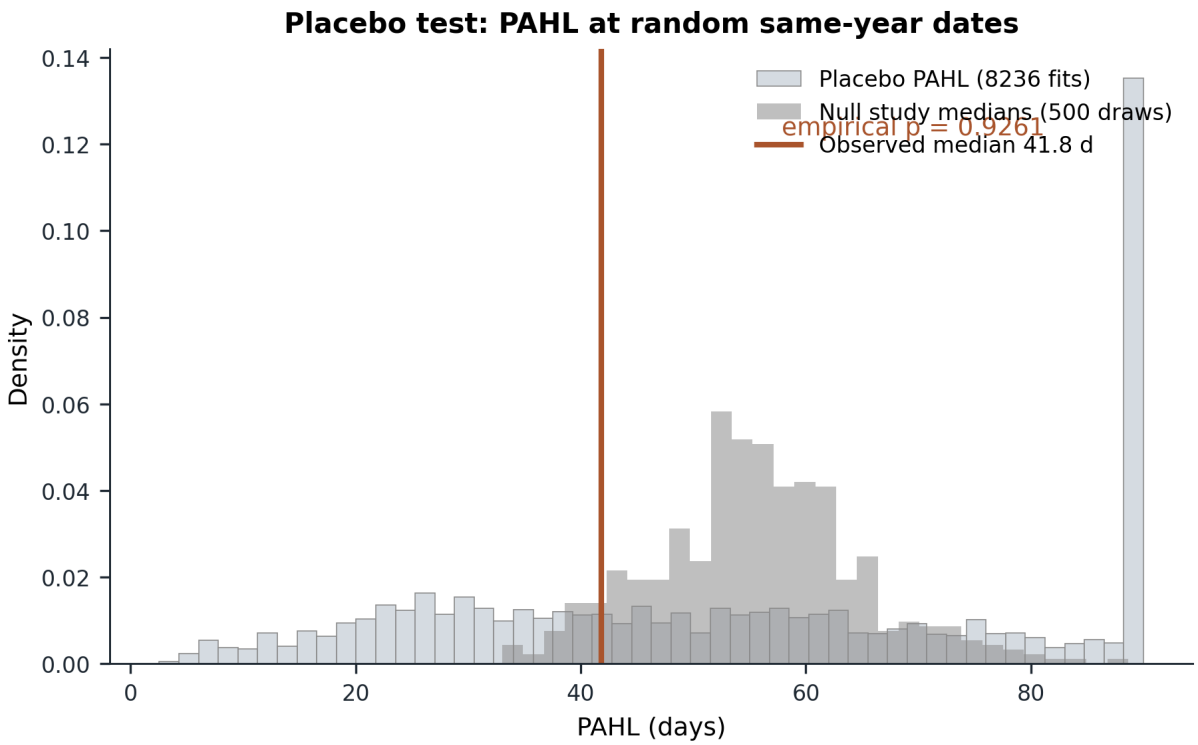


Figure 7. Placebo distribution of PAHL from 8236 fits at random same-year dates (light) against the observed event-level distribution (dark). The observed median of 41.8 days is shorter than the placebo median of 55.3 days, so the ceasefire-specific claim fails.

The spike, unlike the decay, is event-linked

The decay placebo asks whether the post-peak rate is a signature of ceasefires. A different and weaker question is whether the announcement spike is: does attention actually rise more on ceasefire dates than it does on arbitrary dates? The two questions are tested separately because they have different answers.

Drawing the same 500 pseudo-event dates per event from the same calendar year, 13433 of 13500 draws produce a valid peak excess and enter the spike placebo distribution. Observed median peak excess is 0.504 (mean 0.811; n = 27 events). The placebo median is 0.228 (mean 0.367; 95th percentile 1.490). The one-sided empirical p-value for a ceasefire-specific spike is 0.0060, which passes at the raw 5 per cent level.

Two qualifications keep this from being a headline. After Benjamini-Hochberg correction across the ten tests reported in this paper the spike-placebo p-value becomes q = 0.060, which is marginal; the raw result is strong, but the multiplicity correction should be disclosed. And the result is aggregate rather than per-event: only 5 of the 27 events individually beat their own placebo null at p at or below 0.05. The honest statement is that the aggregate announcement-day response is distinguishable from the random-date baseline before correction and not after it.

The spike and the decay are also not two measures of the same thing. Across the 24 events where both statistics exist, the rank correlation between peak excess and PAHL is -0.037 (95 per cent confidence interval -0.432 to 0.392; p = 0.864). That near-zero correlation matters for interpretation: it rules out regression to the mean, the obvious alternative explanation for the failure of the decay placebo, under which larger spikes would mechanically produce shorter half-lives. The decay null is therefore not an artefact of spike magnitude.

The War-Peace Attention Ratio

Across 14 language editions and 125 months, the global median monthly WPAR is 2.65: Wikipedia readers attend to war concepts roughly two and a half times as much as to peace concepts. Cross-language dispersion is substantial, with a median monthly interquartile range of 0.404 across editions. The trend in log WPAR is 0.0186 per year (95 per cent confidence interval -0.0004 to 0.0375; p = 0.055), approximately 1.9 per cent per year, so the imbalance is if anything widening at a rate that conventional thresholds do not quite resolve. Seasonality, measured as the range of month-of-year effects, is 0.120 log units. The highest median ratio is in Russian (4.42) and the lowest in Hebrew (1.64). Figure 4 shows the monthly series by edition.

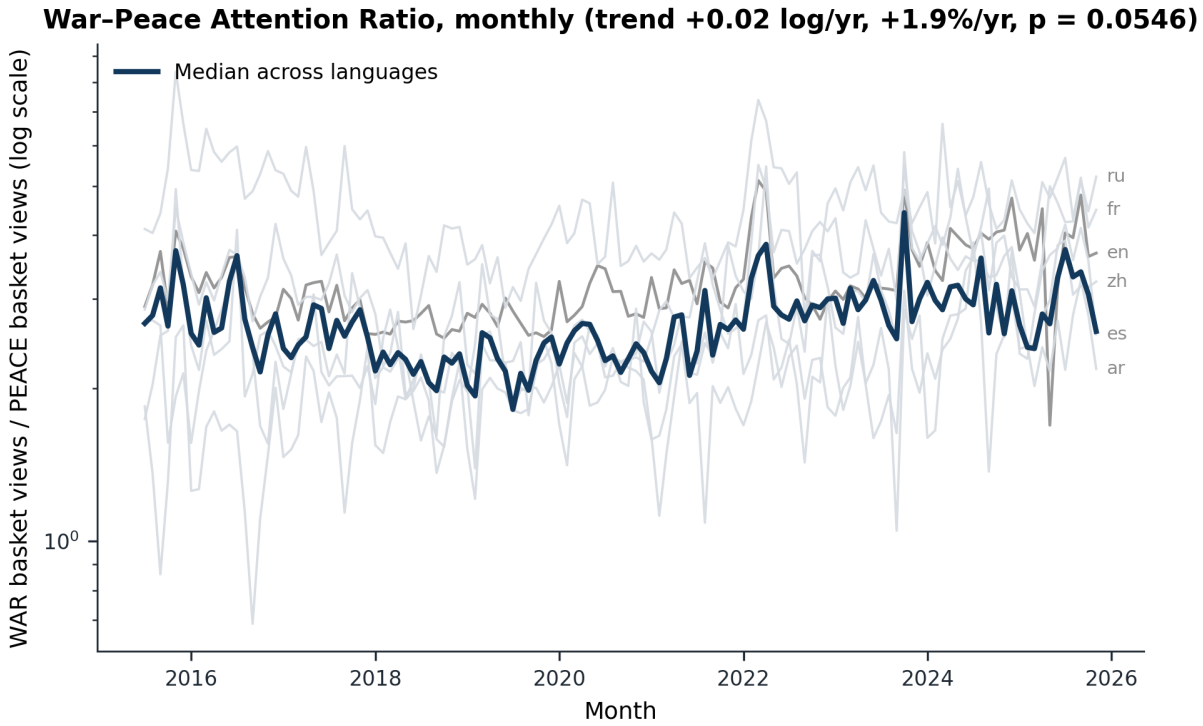


Figure 4. Monthly War-Peace Attention Ratio by Wikipedia language edition, 2015-2025 (14 editions, 125 months). The ratio is war-basket views divided by peace-basket views; the horizontal position of each series reflects its level, and the spread across editions is substantial.

Three cautions apply to the ratio. It is a volume measure, not a sentiment measure: a high ratio means that war concepts are read more, not that peace is less valued. Family composition differs across editions, so a language's level partly reflects how its article set was resolved. And cross-lingual attention comparisons partly reflect connectivity and coverage, as Ruprechter et al. (2023) show, so the ordering of editions should be read as a property of readerships rather than of publics. The trend, which is estimated within edition and aggregated, is less exposed to these composition effects than the levels, and it is the part of the WPAR result we treat as most robust.

Resumption and the exploratory predictive association

Of the 36 events, 33 can be coded for resumption, of which 25 resumed within twelve months, a base rate of 75.8 per cent; 3 cannot be coded because twelve months of follow-up are not yet available. The prediction sample is 23 events, of which 18 resumed and 5 did not, a base rate of 78.3 per cent.

The univariate association runs in the direction opposite to the accountability expectation: a one-standard-deviation longer PAHL is associated with higher odds of resumption, with an odds ratio of 5.32 (95 per cent confidence interval 0.74 to 38.21; p = 0.096). In-sample AUC is 0.800, the repeated cross-validated AUC is 0.775 (SD 0.018 across 200 repeats), and the leave-one-out predictive AUC is 0.667. The Spearman correlation between PAHL and resumption is 0.430 (bootstrap 95 per cent confidence interval 0.115 to 0.690). The adjusted model attenuates the association to an odds ratio of 2.03 (95 per cent confidence interval 0.78 to 5.30; p = 0.150), with an in-sample AUC of 0.967 and a cross-validated AUC of 0.792 (SD 0.054) on twelve parameters. The Cox model for time to resumption gives a log-hazard coefficient of 0.236 (95 per cent confidence interval -0.186 to 0.658; p = 0.273; hazard

ratio 1.266, 95 per cent confidence interval 0.831 to 1.931; n = 23, 18 events). Coefficients are reported in Table 5, and the individual-level pattern is shown in Figures 3 and 8.

Replacing PAHL with the standardised peak excess, the spike model gives an odds ratio of 0.318 (95 per cent confidence interval 0.100 to 1.012; p = 0.052; n = 23, 18 resumed and 5 not; in-sample AUC 0.767; cross-validated AUC 0.695). The negative coefficient means that larger announcement spikes are associated with lower odds of resumption, the opposite direction to the PAHL association. The two statistics are not the same construct, and neither interval excludes one. A median split of PAHL is consistent with the same direction as the PAHL model: the resumption rate is 100.0 per cent above the median (n = 11) and 58.3 per cent below it (n = 12), with a Fisher's exact p of 0.037, although the odds ratio is not finite because the high half contains no non-resumptions. That is a descriptive contrast on 23 events and is not an independent test.

Table 5. Logistic coefficients for the exploratory prediction of resumption, in log-odds with standard errors, odds ratios and 95 per cent intervals. The adjusted model is ridge-penalised; agreement-type and region terms are omitted here and reported in full in the analysis tables.

Model	Term	Coefficient	SE	OR	OR 95 per cent CI	p	n
Univariate	constant	1.910	0.840	-	-	0.023	23
Univariate	PAHL (standardised)	1.672	1.006	5.32	0.74 to 38.21	0.096	23
Adjusted	constant	1.190	0.660	-	-	0.071	23
Adjusted	PAHL (standardised)	0.706	0.490	2.03	0.78 to 5.30	0.150	23
Adjusted	log pre-event fatalities	-0.245	0.419	0.78	0.34 to 1.78	0.558	23

The evidence is contradictory and we report it as such. The logistic p-value of 0.096 and the permutation p-value of 0.033 disagree; the confidence interval on the odds ratio runs from 0.74 to 38.21; five non-resumed events carry the entire contrast; the leave-one-out AUC of 0.667 is weak; and no test survives false-discovery adjustment. On the evidence here the association is exploratory. We do not claim that PAHL predicts resumption, and the disagreement between the tests is the reason rather than an inconvenience.

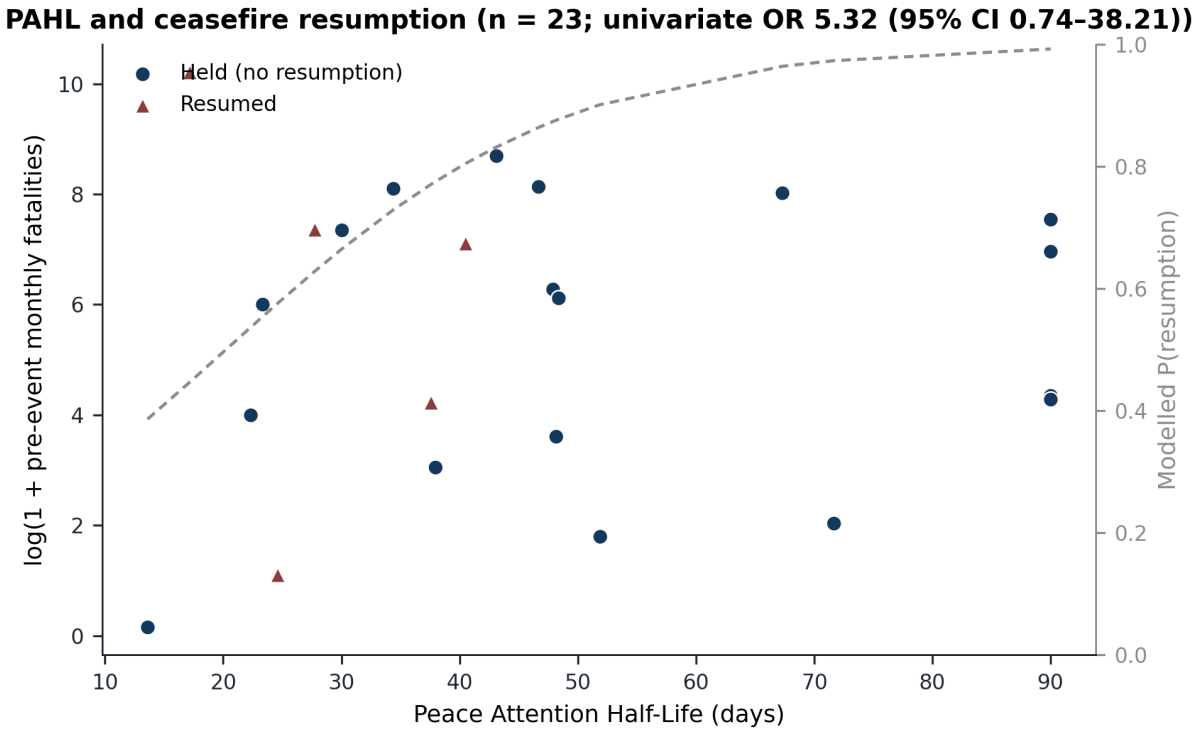


Figure 3. Event-level PAHL against resumption of violence within twelve months (n = 23 events; 18 resumed, 5 did not). The direction is that longer-lived attention accompanies more resumption, the opposite of the enforcement expectation, but the groups overlap heavily.

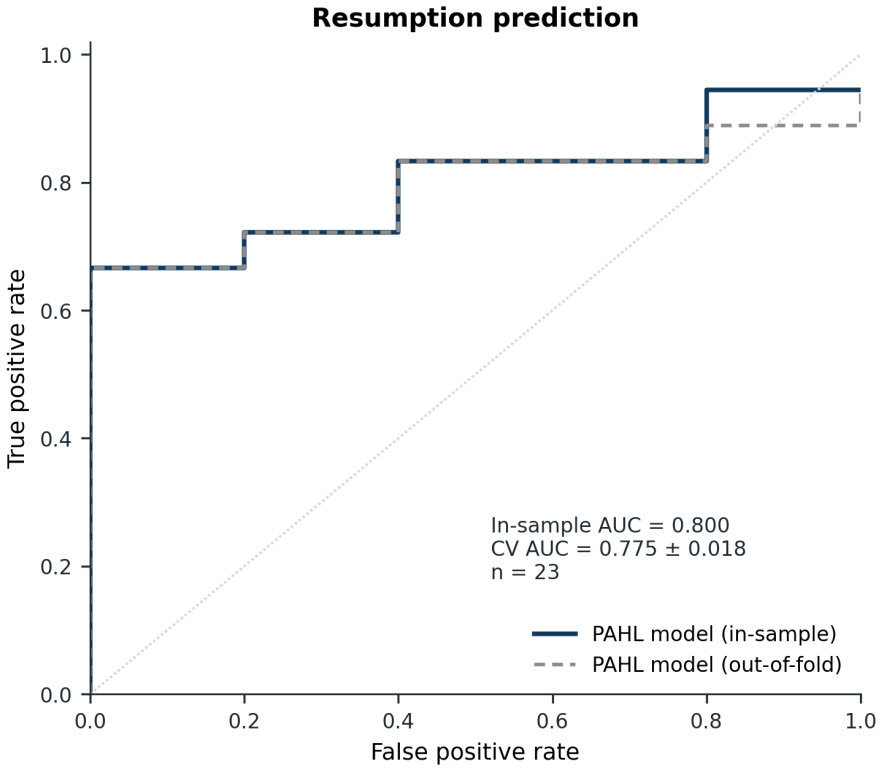


Figure 8. ROC curves for the exploratory prediction models, comparing in-sample and repeated cross-validated performance. The gap between the two is the optimism of the in-sample fit.

Robustness

Decay window. Table 3 reports the median half-life under alternative fitting windows and under a non-parametric half-peak crossing estimate. The level is strongly window-dependent, as expected of a censored decay measure: the median is 15.7 days at a 30-day window, 41.8 days at the primary 90-day window, and 173.5 days at a 270-day window, while the crossing estimate gives 5.5 days. The rank correlation between per-event estimates at the primary window and the alternatives is 0.59 at 30 days and 0.35 at 270 days, so the ordering of events is reasonably stable at short windows and erodes at long ones.

Table 3. Decay-window sensitivity. The crossing estimator is a non-parametric half-peak crossing and does not assume an exponential form.

Window	Median PAHL (days)	IQR	n	95 per cent CI	Spearman with primary
30 days	15.7	8.4 to 25.8	23	10.3 to 21.5	0.59
90 days (primary)	41.8	29.5 to 55.7	24	31.3 to 48.4	1.00
270 days	173.5	130.7 to 230.6	20	136.5 to 209.6	0.35
Crossing	5.5	3.8 to 10.5	24	4.0 to 8.0	-

Baseline definition. Fitting on excess over the rolling baseline rather than the pre-event baseline gives a median of 45.4 days (interquartile range 38.4 to 69.9; n = 24) against 41.8 days for the primary definition, a shift of about a week that does not change the interpretation.

Resumption threshold. Table 4 reports the association at alternative thresholds. At 10 per cent the base rate is 0.848 and the odds ratio is 5.32 (0.74 to 38.21; p = 0.096; cross-validated AUC 0.775); at 25 per cent the base rate is 0.758 and the odds ratio is 5.32; at 50 per cent the base rate is 0.727, the odds ratio falls to 1.66 (0.57 to 4.88; p = 0.355) and the cross-validated AUC to 0.587. The association is not monotone in the threshold, and the result should not be described as threshold-robust.

Table 4. Resumption-threshold sensitivity. The 10 and 25 per cent definitions code the prediction sample identically; the 50 per cent definition attenuates the association.

Threshold	Coded	Base rate	OR	95 per cent CI	p	In-sample AUC	CV AUC
0.10	33	0.848	5.32	0.74 to 38.21	0.096	0.800	0.775
0.25 (primary)	33	0.758	5.32	0.74 to 38.21	0.096	0.800	0.775
0.50	33	0.727	1.66	0.57 to 4.88	0.355	0.672	0.587

Leave-one-out and permutation. Deleting each event in turn moves the cross-event median by at most 1.31 days (leave-one-out medians 40.5 to 43.1; SD 1.33; full-sample median 41.8), so the measurement summary is stable. The prediction is not: leave-one-out predictive AUC is 0.667, and refitted in-sample AUCs on the remaining events range from 0.764 to 0.847 with a median of 0.788. Permuting the resumption outcome 1000 times gives a two-sided permutation p-value of 0.033 for the standardised PAHL coefficient (observed 0.965; null central 95 per cent interval -0.826 to 0.959); the Spearman permutation p-value is 0.052.

Language and mapping subsets. Restricting to English-language series gives a median of 43.4 days (interquartile range 14.2 to 76.2; n = 8; Spearman with the multilingual estimate 0.952), while the multilingual-only subset of events with at least two editions gives 41.8 days (interquartile range 28.3 to 62.5; n = 18). The headline does not hinge on English. Restricting to exact and high-confidence mappings gives 40.5 days (26.2 to 50.0; n = 19).

Multiplicity. Across the ten main hypothesis tests, Benjamini-Hochberg adjustment at a 5 per cent false-discovery rate leaves 0 significant. The permutation p-value of 0.033 for the PAHL association becomes 0.109 after adjustment, and the spike-placebo p-value of 0.006 becomes 0.060. The exploratory models are therefore not presented as confirmatory either way, and a reader who applies a multiplicity correction to our p-values should conclude that we have not established an effect.

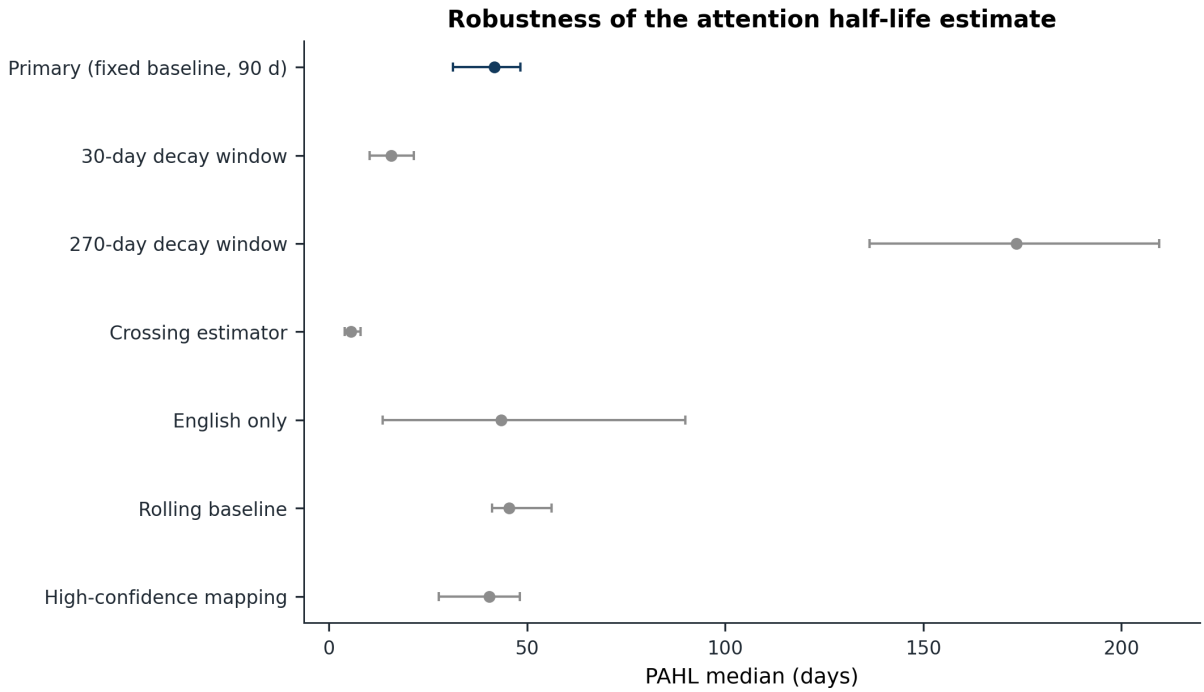


Figure 10. Sensitivity of the PAHL median and of the predictive AUC to alternative decay windows, resumption thresholds and leave-one-out deletion.

6. Discussion

What was measured

This paper asked three questions: how fast attention to a ceasefire decays, whether attention to peace decays differently from attention to war, and whether the decay rate predicts whether the ceasefire holds. The answers are uneven, and the unevenness is the result.

The first answer has two parts. Attention to a ceasefire announcement spikes on the day of the announcement and reaches half its peak within two weeks; the per-event decay of that spike fits an exponential with a median half-life of about six weeks. But the same procedure applied to arbitrary dates in the same years produces a longer median half-life of 55.3 days, so the measured decay is not a signature of ceasefires. Our own pre-specified falsification condition is therefore met, and the honest interpretation is that PAHL describes the attrition schedule of attention to conflict content rather than a ceasefire-specific

response. This is a negative result for the measurement claim and, we think, an informative one for the attention-decay literature: the decay laws estimated on cultural items (Candia et al. 2019) transfer to conflict content in the sense that a half-life exists and is estimable, but the event class does not change the rate. A study that had not run the placebo test would have reported a six-week half-life for peace as a discovery.

The failure of the decay placebo is not a regression-to-the-mean artefact. The rank correlation between spike magnitude and half-life is -0.037 ($p = 0.864$; $n = 24$), which is essentially zero: larger announcement spikes do not mechanically produce shorter half-lives in these data, so the decay null is a property of the decay rather than of the spike.

The attention deficit is one of volume, not persistence

Second, the War-Peace Attention Ratio is a large and stable descriptive fact: readers of Wikipedia attend to war concepts roughly two and a half times as much as to peace concepts, and the imbalance is highest in Russian at 4.42 and lowest in Hebrew at 1.64. The basket-level half-lives point the other way from the project's initial expectation: peace concepts decay more slowly at 81.6 days against 65.3 days for war concepts, although the paired difference is small and its interval spans zero. The attention deficit, in other words, is a deficit in the volume of attention, not in its persistence. War attracts more attention and sheds it faster; peace attracts less and keeps it longer.

This reversal matters for how the result should be framed publicly. The intuitive story, that the world forgets peace faster than war, is not what the data show. What the data show is that war is more newsworthy and more volatile, which is consistent with the news-values account in which violence supplies the drama that makes an event reportable (Galtung and Ruge 1965). The ratio measures the volume of the imbalance, and the basket comparison shows that the persistence of attention does not follow the volume.

The predictive signal, and why we do not claim it

Third, the decay rate is associated with resumption in the direction opposite to the accountability mechanism: longer-lived attention accompanies more violence resumption, not less. If that association were real, it would suggest that attention persists when a conflict remains newsworthy, so that the decay rate is a marker of ongoing volatility rather than a protective resource. That reading is consistent with the salience mechanism and with the observation that the decay of attention is generic across event classes. It is, however, not established. The odds ratio is estimated on 23 events with 5 non-resumed cases; its interval runs from 0.74 to 38.21; the permutation test and the model-based test disagree; the 50 per cent threshold attenuates the association sharply; and no test survives false-discovery adjustment. We report the direction because it is stable across specifications and theoretically interpretable, and we decline to report it as a finding.

The two exploratory signals point in opposite directions, which is worth stating plainly because it is the honest state of the evidence. A longer half-life is associated with higher odds of resumption (odds ratio 5.32; $p = 0.096$), while a larger announcement spike is associated with lower odds (odds ratio 0.318; $p = 0.052$). Slower decay and a larger announcement response are not the same construct. If both associations were real, they would describe a conflict that draws the largest crowd at the announcement and then holds it longest, which on the salience reading is a conflict that remains newsworthy because it remains violent.

Neither association survives false-discovery correction, both rest on five non-resumed events, and we present the opposition as an open question for a larger sample rather than as a finding.

What would be needed to make the predictive claim credible is not a better estimator but a better design: a larger event set drawn from the population of ceasefires with documented selection rules, event-specific attention from more than one platform, a placebo matched on local violence intensity rather than on calendar year, pre-registration of the thresholds, and hierarchical shrinkage across events and editions. None of these is beyond reach. The population-level ceasefire dataset of Clayton et al. (2022) supplies the first, and combining it with the measurement pipeline released with this paper is the obvious next step.

Implications for early warning

The paper was framed as a test of a candidate early-warning lead indicator, and the result of that test is negative as implemented. Attention decay is observable at announcement, in public data, at low cost, across languages, including in conflicts without peacekeeping missions or coverage in the largest news markets, and those properties are genuinely attractive for early warning. But an indicator that does not separate the event from the background does not add information to a forecast, and an association that cannot be estimated with a stable sign is not a usable input. The honest conclusion for the early-warning literature is that attention to a peace process, at least as measured through encyclopaedic pageviews, does not yet carry usable information about whether the process holds. That is a useful negative result: it narrows the search, and it is exactly the kind of result that the pre-specified falsification conditions were designed to produce.

7. Limitations

Proxy measurement. Wikipedia pageviews are an attention proxy skewed toward literate, online populations, and the multilingual median mitigates but does not eliminate the dominance of the largest editions. Pageviews measure engagement with an encyclopaedia article, not support for a policy, and Ruprechter et al. (2023) show that event attention is skewed by wealth and region, so cross-lingual differences are partly differences in connectivity. Nothing here measures public opinion.

Small samples and selection. Twenty-four events have a half-life and 23 enter the prediction sample; 12 events produce no estimate, and the loss is not random, because the panel under-represents sudden new wars whose articles did not exist before the announcement. The sample departs from the population-level ceasefire dataset for reasons of measurability, so the estimates describe the cases we could measure rather than ceasefires in general.

Event-to-article mapping. Attention is measured on the conflict article that the mapping step judged to cover the theatre, and that judgement is heuristic; every mapping, its method and its confidence are reported in Table 1. Restricting to exact and high-confidence mappings gives a median of 40.5 days (n = 19), close to the headline. The remainder includes at least two events whose best available article covers a related rather than the same conflict: the May 2025 India-Pakistan ceasefire is measured against the article on insurgency in Khyber Pakhtunkhwa, and the 2015 Myanmar ceasefire against an article whose subject is the civil war that began in 2021. Where the correct article did not exist before the announcement the event yields no half-life and is excluded, but two events with usable fits rest on mappings we regard as weak, and the event-level half-lives should be treated as indicative rather than exact.

No causal identification. Attention and violence are jointly determined; the adjusted model controls for pre-event fatalities but cannot separate attention from the conflict dynamics that generate both. Nothing here identifies an effect of attention on survival, and the sign of the association is compatible with attention being a consequence of violence rather than a cause of restraint.

Measurement construction. Pageview series are sums across page moves and title variants; articles created mid-period cover only part of the window; views to redirect titles are not aggregated; weekly structure is handled by smoothing; and the baseline choice matters, with the rolling-baseline variant giving 45.4 days against the primary 41.8. The estimator is a single exponential with no offset, so a slow cultural component is not identified and residual drift can bias lambda. The ninety-day window censors 0.233 of fits, and the level of the summary depends on the horizon, as the comparison from 15.7 days at a 30-day window to 173.5 days at a 270-day window shows.

Outcome coding. Resumption uses the 25 per cent, six-month, two-consecutive-month rule on country-level UCDP GED fatalities; Gaza events are joined to Israeli fatalities through the documented proxy; the twelve-month follow-up window begins with the event month, so a late-month announcement can have pre-announcement violence counted in its first post month; events at the edge of fatality coverage in 2024 and 2025 cannot be coded; and three events lack a full twelve-month follow-up. The 50 per cent threshold attenuates the predictive association materially, so the outcome definition is load-bearing for the one positive signal we observe.

Placebo validity. Random same-year dates are an imperfect null: in conflict-heavy years they inherit other escalations, which inflates the null half-life and makes the test conservative against the ceasefire-specific claim. Even so, the observed decay is not distinguishable from the null and we do not have a post hoc rescue. The absence of evidence for event-specific decay is not proof that no ceasefire-specific component exists; it is evidence that this design cannot detect one.

Multiplicity and researcher degrees of freedom. The paper reports ten main hypothesis tests plus several descriptive contrasts, and no test survives false-discovery adjustment. The two exploratory signals also point in opposite directions, a longer half-life being associated with higher odds of resumption and a larger spike with lower odds, and neither survives correction; we report the opposition as the open question it is. Language editions, windows, thresholds and subsets create degrees of freedom; all the choices reported here were pre-specified in the project design and all are reported, but the pre-specification is a design document rather than a registered report.

8. Conclusion

We defined the Peace Attention Half-Life and estimated it for 24 of 36 verified ceasefires across 14 Wikipedia language editions, alongside a cross-lingual War-Peace Attention Ratio and an exploratory test of whether the decay rate predicts the resumption of violence. Attention to a ceasefire spikes on the announcement day, halves within two weeks, and decays with a median half-life of 41.8 days. But that rate is statistically indistinguishable from the rate at which attention to arbitrary dates decays in the same years, so we interpret it as the attrition schedule of attention to conflict content and not as a ceasefire-specific response. Wikipedia readers attend to war concepts 2.65 times as much as to peace concepts, with the ratio highest in Russian at 4.42 and lowest in Hebrew at 1.64 and a slow upward trend, while the persistence of attention shows no evidence of being shorter for peace. A longer half-life was

associated with higher odds of resumption, the opposite of the enforcement expectation, but the association is fragile, threshold-sensitive and not significant after multiplicity adjustment. The announcement-day rise is an aggregate response at the raw level ($p = 0.006$), though marginal after correction ($q = 0.060$), and the two exploratory signals point in opposite directions without either surviving adjustment.

The contribution is a measurement, a cross-lingual descriptive fact, and a null: the instrument is reproducible, the datasets are frozen, and the finding that attention decay is generic is reported rather than explained away.

Data and code availability

All code, data and figures are available at https://github.com/AhmedJaffar/half-life-of-peace. The repository contains the retrieval pipeline, which is cached and resumable; the frozen processed datasets with recorded retrieval dates and SHA-256 hashes in `analysis/manifest.json`; the analysis code; the figure code; the machine-readable results file `analysis/results.json`; the nine analysis tables; and the build script for this manuscript. Code is MIT-licensed, and data and manuscript are CC BY 4.0. Pageview data derive from the Wikimedia Analytics API, fatality data from UCDP GED v26.1, and the curated ceasefire table is released with per-row source URLs.

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